FUNDAMENTALS OF DATA SCIENCE

IMP 8-MARK QUESTIONS

UNIT I – INTRODUCTION TO DATA SCIENCE

1. Define Data Science. Explain its scope and importance in various fields.
2. Explain the history and evolution of Data Science.
3. Explain the major applications of Data Science.
4. Explain the Data Science Lifecycle with a neat diagram and describe each stage.
5. Compare arrays, lists, matrices, and Data Frames with suitable examples.
6. Explain the important tools and technologies used in Data Science.

UNIT II – DATA COLLECTION AND PRE - PROCESSING

1. Explain different methods of data collection with suitable examples.
2. What is data pre-processing? Explain the major steps involved in data pre-processing.
3. Explain data cleaning techniques with examples.
4. Explain different techniques for handling missing values with an example.
5. Explain normalization and standardization. Compare them with suitable examples and formulas.
6. What is feature engineering? Explain different feature engineering techniques with examples.
7. Explain feature scaling with suitable example.
8. Explain data integration and data aggregation with suitable example.

UNIT III – DATA LOADING, STORAGE AND FILE FORMATS

1. Describe the methods used to read and write data in text files.
2. Explain how a large text file can be read in pieces.
3. Demonstrate how to work with delimited data formats using a suitable example.
4. Describe the structure and uses of JSON data format with a suitable example.
5. Classify the main features of HTML and XML data format with suitable examples.
6. Demonstrate the process of collecting data from a web page using web scraping.
7. Distinguish between Binary data formats such as HDF5 and traditional file formats. Indicate their advantages and suitable use cases
8. Apply NoSQL Mongo DB CRUD operations to store and retrieve by taking suitable data as an example.

UNIT IV – EXPLORATORY DATA ANALYSIS

1. Describe the commonly used measures of descriptive statistics with suitable examples.
2. Demonstrate different data visualization techniques used in Exploratory Data Analysis with suitable examples.
3. Analyze the correlation and covariance techniques by using suitable example.
4. Differentiate between Univariate, Bivariate and Multivariate analysis with suitable examples.
5. Analyze a dataset to identify possible Anomalies and Pattern Detection.

UNIT V – DATA SCIENCE TOOLS

1. Describe Neo4j and its basic features as a graph database.
2. Demonstrate basic Cypher queries with suitable examples.
3. Describe the role of NLTK in text mining and analytics with suitable examples.
4. Analyze the differences between Neo4j and SQLite based on their data models and uses.
5. Apply different tools used for text analytics and review the use of SQLite in managing text-based data.